

course, not new; the first eukaryotes had long ago invented a strategy—phagotrophy—for killing life. But this worked only on level-one (single-cellular) life; level-two multicellular blobs were far too big to engulf into a single cell. And so early animals evolved internal digestion as a strategy for eating level-two life. Animals uniquely form little stomachs where they trap prey, secrete enzymes, and digest them.

In fact, the formation of an inner cavity for digestion may have been the defining feature of these early animals. Practically every animal alive today develops in the same three initial steps. From a single-celled fertilized egg, a hollow sphere (a blastula) forms; this then folds inward to make a cavity, a little “stomach” (a gastrula). This is true of human embryos as much as jellyfish embryos. While every animal develops this way, no other kingdom of life does this. This provides a glaring hint as to the evolutionary template on which all animals derive: we form stomachs for ingesting food. All animals that engage in such gastrulation also have neurons and muscles and seem to derive from a common neuron-enabled animal ancestor. Gastrulation, neurons, and muscles are the three inseparable features that bind all animals together and separate animals from all other kingdoms of life.

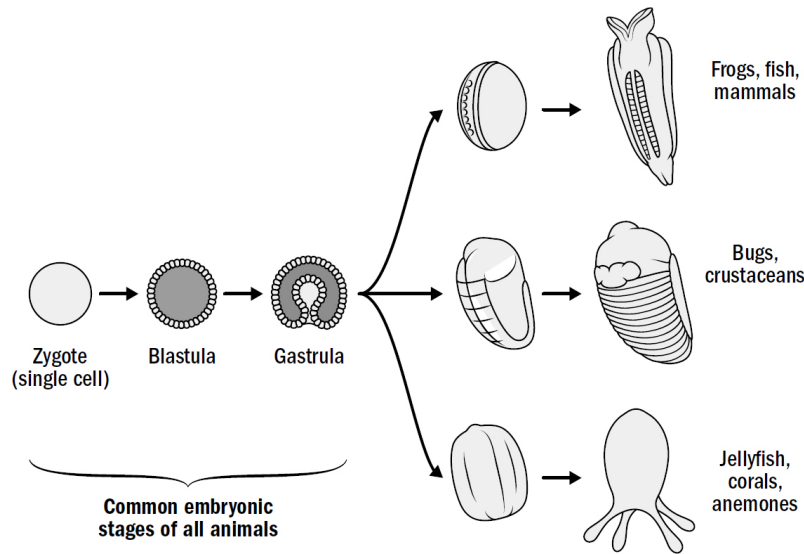


Figure 1.6: Shared developmental stages for all animals

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Some have gone so far as to argue that the grandmother of animals was literally a little gastrula-shaped creature with neurons. But this is a scientific arena full of controversy—just because all animals develop this way doesn't mean that they ever truly lived in this form.

Another interpretation, supported by fossils, suggests that the first animals may have been similar to today's corals. To the naked eye, a coral is so simple that it doesn't look all that different from a fungus or a plant ([figure 1.8](#)). Only when examining its biology closely would you see the presence of the animal template: a stomach, muscles, and neurons. A coral is actually a colony of independent organisms called coral polyps. A coral polyp is, in some sense, literally *only* a stomach with neurons and muscles. They have little tentacles that float in the water, waiting for small organisms to swim toward them. When food touches one of the tips of these tentacles, they rapidly contract, pulling the prey into the stomach cavity where it is digested. Neurons on the tips of these tentacles detect food and trigger a cascade of signaling through a web of other neurons that generates a coordinated relaxing and contracting of different muscles.

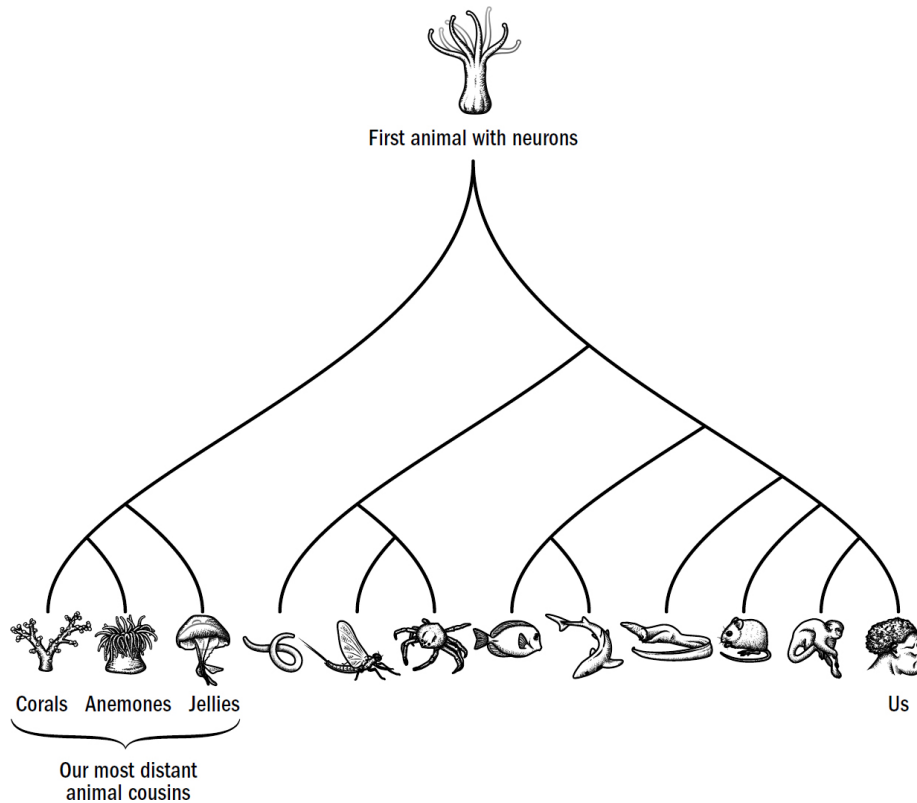


Figure 1.7: Tree of neuron-enabled animals

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This coral reflex was not the first or the only way in which multicellular life sensed and responded to the world. Plants and fungi do this just fine without neurons or muscles; plants can orient their leaves toward the sun, and fungi can orient their growth in the direction of food. But still, in the ancient sea at the dawn of multicellularity, this reflex would have been revolutionary, not because it was the first time multicellular life sensed or moved but because it was the first time it sensed and moved with *speed* and *specificity*. The movement of plants and fungi takes hours to days; the movement of coral takes seconds.* The movement of plants and fungi is clumsy and inexact; the movement of coral is comparatively very specific—the grasping of prey, opening of the mouth, pulling into the stomach, and closing of the mouth all require a well-timed and accurate orchestration of relaxing some muscles while contracting others. And *this* is why fungi don't have neurons and animals do. Although both are large multicellular organisms that feed on other life, only the animal-survival strategy of

killing level-two multicellular life requires fast and specific reflexes.* The original purpose of neurons and muscles may have been the simple and inglorious task of swallowing.

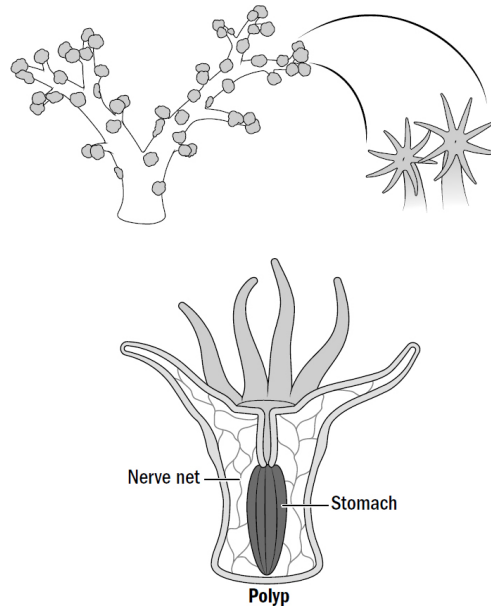


Figure 1.8: Soft coral as a model for early animal life

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Edgar Adrian's Three Discoveries and the Universal Features of Neurons

The scientific journey by which we have come to understand how neurons work has been long and full of false starts and wrong turns. As far back as the famous ancient Greek physician Hippocrates in 400 BCE, it has been known that animals contain a system of stringy material later called *nerves* (from the Latin word *nervus*, meaning “sinew”), flowing to and from the brain, and that this system is the medium by which muscles are controlled and sensations are perceived. They knew this through horrifying experiments of severing spinal cords and clamping nerves in living pigs and other livestock. But the ancient Greeks wrongly concluded that it was “animal spirits” that flowed through these nerves. It would be many centuries before this error was corrected. More than two thousand years

later, even the great Isaac Newton would speculate, incorrectly, that nerves communicated via vibrations flowing through a nerve fluid he called *ether*. It wasn't until the late 1700s that scientists discovered that flowing through the nervous system was not *ether* but *electricity*. Electrical current applied to a nerve will flow toward downstream muscles and cause them to contract.

But many errors remained. At the time, it was thought that the nervous system was made up of a single web of nerves, analogous to the circulatory system's homogenous web of blood vessels. It wasn't until the end of the nineteenth century, with more advanced microscopes and staining techniques, that scientists discovered that the nervous system was made up of independent cells—neurons—that, although wired together, are separate and generate their own signals. This also revealed that electrical signals flow only in a single direction within a neuron, from one part that takes inputs, called *dendrites*, to another that sends electrical output, called the *axon*. This output flows to other neurons or other types of cells (such as muscle cells) to activate them.

In the early 1920s, a young English neurologist by the name of Edgar Adrian returned to Cambridge University after a long stint of medical service during World War I. Adrian, like many researchers at the time, was interested in electrically recording neurons to decipher how and what they communicated. The problem had always been that electrical recording devices were too large and crude to record the activity of a single neuron and hence always produced a noisy mess of signals from multiple neurons. Adrian and his collaborators were the first to find a technical solution to this problem, inventing the field of single-neuron electrophysiology. This gave scientists, for the first time, a view into the language of individual neurons. The subsequent three discoveries earned Adrian the Nobel Prize.

The first discovery was that neurons don't send electrical signals in the form of a continuous ebbing and flowing but rather in all-or-nothing responses, also called spikes or action potentials. A neuron is either on or off; there is no in between. In other words, neurons act less like an electric power line with a constant flow of electricity and more like an electric telegraph cable, with patterns of electrical clicks and pauses. Adrian himself noted the similarity between neural spikes and Morse code.

This discovery of spikes presented a puzzle for Adrian. You can clearly perceive *stimulus strengths* in your senses—you can discriminate between

different volumes of sound, brightness of light, potency of smells, severity of pain. How could a simple binary signal that was either on or off communicate a numerical value, such as the graded strengths of stimuli? The realization that the language of neurons was spikes didn't tell researchers much about what a sequence of spikes *meant*. Morse code is, well, a code—it is an efficient trick to store and transmit information across a single electrical wire. Adrian was the first scientist to use the word *information* to refer to the signals of neurons, and he devised a simple experiment to try to decode them.

Adrian took a muscle from the neck of a deceased frog and attached a recording device to a single stretch-sensing neuron in the muscle. Such neurons have receptors that are stimulated when muscles are stretched. Adrian then attached various weights to the muscle. The question was: How would the responses of these stretch-sensing neurons change based on the *weight* placed on the muscle?

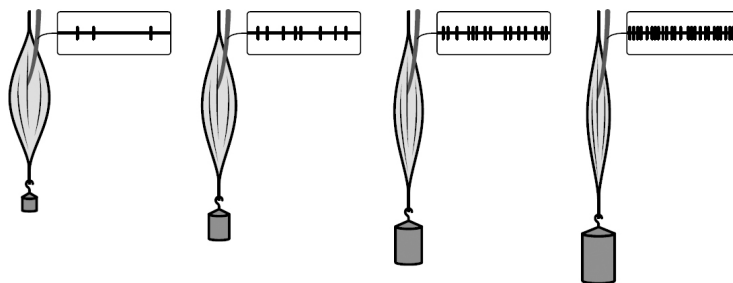


Figure 1.9: Adrian charted the relationship between weight and the number of spikes per second (i.e., the *spike rate*, or *firing rate*) elicited in these stretch neurons.

Original art by Rebecca Gelernter

It turned out the spikes were identical in all cases; the only difference was *how many* spikes were fired. The heavier the weight, the higher the frequency of spikes ([figure 1.9](#)). This was Adrian's second discovery, what is now known as *rate coding*. The idea is that neurons encode information in the rate that they fire spikes, not in the shape or magnitude of the spike itself. Since Adrian's initial work, such rate coding has been found in neurons across the animal kingdom, from jellyfish to humans. Touch-sensitive neurons encode *pressure* in their firing rate; photosensitive neurons encode *contrast* in their firing rate; smell-sensitive neurons encode

concentration in their firing rate. The neural code for movement is also a firing rate: the faster the spikes of the neurons that stimulate muscles, the greater the contraction force of the muscles. This is how you are able to delicately pet your dog and also pick up fifty-pound weights—if you couldn't modulate the strength of muscle contractions, you wouldn't be very pleasant to be around.

Adrian's third discovery was the most surprising of all. There is a problem with trying to translate natural variables, such as the pressure of touch or the brightness of light into this language of rate codes. The problem is this: these natural variables have a *massively larger* range than can be encoded in the firing rate of a neuron.

Take vision, for example. What you don't realize (because your sensory machinery abstracts it away) is that the luminance of light is varying astronomically all around you. The amount of light entering your eyes when you look at a white piece of paper is one million times greater if you are in sunlight than if you are in moonlight.* In fact, the black letters of a page in sunlight are thirty times brighter than the white of a page in moonlight!

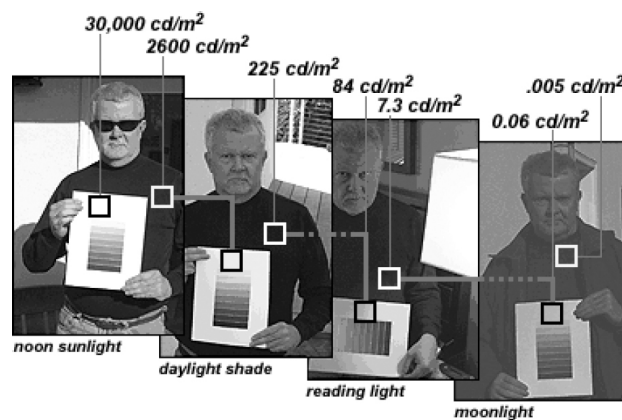


Figure 1.10: The broad spectrum of stimulus strengths

Figure made by B. MacEvoy, 2015. Used with permission (personal correspondence).

This is not just a feature of light; all sensory modalities, from smell to touch to sound, require the discrimination of hugely varying natural variables. This wouldn't necessarily be a problem except for a big limitation of neurons—for a variety of biochemical reasons, it is simply impossible

for a neuron to fire faster than around five hundred spikes per second. This means that a neuron needs to encode a range of natural variables that can vary by a factor of over a million all within a firing rate ranging only from 0 to 500 spikes per second. This could be reasonably called the “squishing problem”: neurons have to *squish* this huge range of natural variables into a comparably minuscule range of firing rates.

This makes rate coding, on its own, untenable. Neurons simply cannot directly encode such a wide range of natural variables in such a small range of firing rates without losing a huge amount of precision. The resulting imprecision would make it impossible to read inside, detect subtle smells, or notice a soft touch.

It turns out that neurons have a clever solution to this problem. Neurons do not have a fixed relationship between natural variables and firing rates. Instead, neurons are always adapting their firing rates to their environment; they are constantly remapping the relationship between variables in the natural world and the language of firing rates. The term neuroscientists use to describe this observation is *adaptation*; this was Adrian’s third discovery.

In Adrian’s frog muscle experiments, a neuron might fire one hundred spikes in response to a certain weight. But after this first exposure, the neuron quickly adapts; if you apply the same weight shortly after, it might elicit only eighty spikes. And as you keep doing this, the number of spikes continues to decline. This applies in many neurons throughout the brains of animals—the stronger the stimuli, the greater the change in the neural threshold for spiking. In some sense, neurons are more a measurement of relative changes in stimulus strengths, signaling how much the strength of a stimulus changed relative to its baseline as opposed to signaling the absolute value of the stimulus.

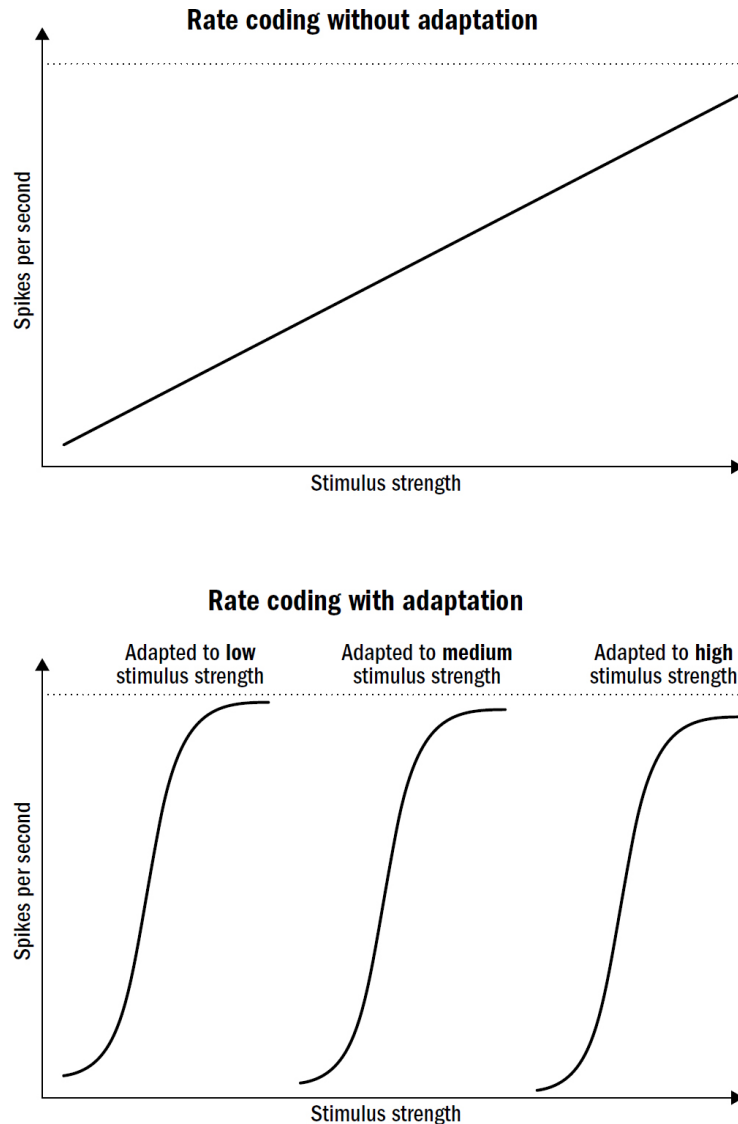


Figure 1.11

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Here's the beauty: Adaptation solves the squishing problem. Adaptation enables neurons to precisely encode a broad range of stimulus strengths despite a limited range of firing rates. The stronger a stimulus is, the more strength will be required to get the neuron to respond similarly next time. The weaker a stimulus is, the more sensitive neurons become.

The late nineteenth and early twentieth centuries were filled with rich discoveries about the inner workings of neurons. A long roster of giants in neuroscience emerged in this period, leading to a platter of Nobel Prizes,

not only Edgar Adrian, but also Santiago Ramón y Cajal, Charles Sherrington, Henry Dale, John Eccles, and others. One important discovery was that nerve impulses pass from one neuron to another by way of synapses, which are microscopic gaps between neurons. Spikes in the input neuron trigger the release of chemicals called neurotransmitters, which travel across the synapse over the course of nanoseconds, attach to a bunch of protein receptors, trigger ions to flow into the target neuron, and thereby change its charge. While neural communication *within* a neuron is electrical, *across* neurons, it is chemical.*

In the 1950s, John Eccles discovered that neurons come in two main varieties: excitatory neurons and inhibitory neurons. Excitatory neurons release neurotransmitters that *excite* neurons they connect to, while inhibitory neurons release neurotransmitters that *inhibit* neurons they connect to. In other words, excitatory neurons *trigger* spikes in other neurons, while inhibitory neurons *suppress* spikes in other neurons.

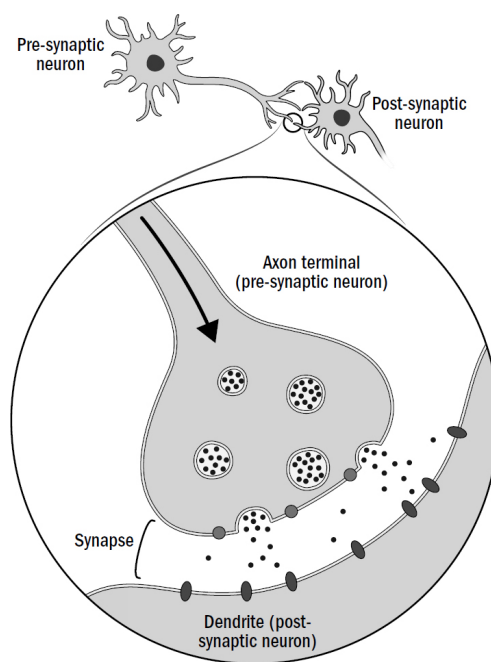


Figure 1.12

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These features of neurons—all-or-nothing spikes, rate coding, adaptation, and chemical synapses with excitatory and inhibitory neurotransmitters—are universal across all animals, even in animals that

have no brain, such as coral polyps and jellyfish. Why do all neurons share these features? If early animals were, in fact, like today's corals and anemones, then these aspects of neurons enabled ancient animals to successfully respond to their environment with speed and specificity, something that had become necessary to actively capture and kill level-two multicellular life. All-or-nothing electrical spikes triggered rapid and orchestrated reflexive movements so animals could catch prey in response to even the subtlest of touches or smells. Rate coding enabled animals to modify their responses based on the strengths of a touch or smell. Adaptation enabled animals to adjust the sensory threshold for when spikes are generated, allowing them to be highly sensitive to even the subtlest of touches or smells while also preventing overstimulation at higher strengths of stimuli.

What about inhibitory neurons? Why did they evolve? Consider the simple task of a coral polyp opening or closing its mouth. For its mouth to open, one set of muscles must contract and another must relax. And the converse for closing its mouth. The existence of both excitatory and inhibitory neurons enabled the first neural circuits to implement a form of logic required for reflexes to work. They can enforce the rule of “do this, not that,” which was perhaps the first glimmer of intellect to emerge from circuits of neurons. “Do this, not that” logic was not new—such logic already existed in the protein cascades of single cells. But this ability was recapitulated in the medium of neurons, which made such logic newly possible on the scale of level-three multicellular life. Inhibitory neurons enabled the necessary inner logic for catch-and-swallow reflexes to work.

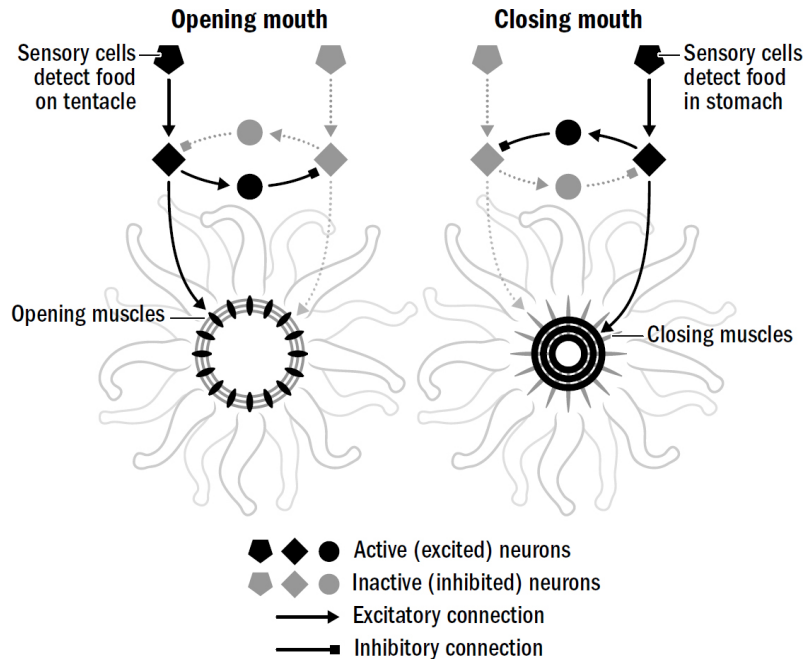


Figure 1.13: The first neural circuit

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While the first animals, whether gastrula-like or polyp-like creatures, clearly had neurons, they had no brain. Like today's coral polyps and jellyfish, their nervous system was what scientists call a nerve net: a distributed web of independent neural circuits implementing their own independent reflexes.

But with the evolutionary feedback loop of predator-prey in full force, with the animal niche of active hunting, and the building blocks of neurons in place, it was only a matter of time before evolution stumbled on [breakthrough #1](#), which led to rewiring nerve nets into brains. It is here where our story truly begins, but it doesn't begin in the way you might expect.